Inverse Design and Simulation Validation of Seismic Metamaterials

Author

Yucheng Ni kevin.ni22@imperial.ac.uk

Abstract

Seismic metamaterials offer a novel means to control destructive ground motion, but their design is hindered by the inefficiency of conventional forward approaches. This study introduces an inverse design framework that integrates finite element simulations with deep learning. Bandgaps of periodic soil–concrete unit cells were computed using FEM in COMSOL, forming the basis for training. A variational autoencoder (VAE) was used to compress high-dimensional geometries into latent vectors, while a multilayer perceptron (MLP) surrogate model enabled rapid bandgap prediction. An inverse network then optimized latent vectors toward target bandgaps, with decoded structures revalidated through FEM. Three significant earthquake case studies demonstrated a close agreement between the designed and target bandgaps. The limitations of the current framework and possible future directions are also discussed.

Keywords —— *Metamaterial; Deep learning; Seismic Wave; bandgap; finite element method*

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I. Introduction and Motivation

Seismic activity has long posed significant threats to human society, causing an estimated 1.87 million deaths in the 20th century [1] and, between 1990 and 2010, an average of 2,052 fatalities per major event [1]. In the United States alone, annual earthquake-related losses are approximately USD 14.7 billion, including indirect damages [2]. These figures underscore the urgent need for innovative mitigation strategies beyond conventional methods. Traditional approaches rely on reinforced materials or dampers to resist seismic wave energy; while partially effective, they inherently oppose seismic waves with the building itself, leading to inevitable degradation [3].

Metamaterials—engineered within the framework of left-handed materials and designed to exhibit properties absent in nature—have recently emerged as a promising alternative. Constructed by periodically or specially designed units, metamaterials were first proposed by Veselago in 1968 [4] and experimentally realized by Pendry and Smith et al. [5]. They have since expanded beyond electromagnetic applications to include acoustic and mechanical waves [6]. Because their properties depend primarily on geometry, metamaterials can be tailored to control wave propagation for applications such as cloaking [7], super-resolution imaging [8], vibration isolation [9].

However, metamaterial's performance varies with geometry, and conventional forward design is constrained by computational cost and limited design space [10]. To address this, inverse-design methods using deep learning have been applied [11], allowing efficient exploration of design space through gradient-based optimization. These methods have identified phononic crystal and acoustic metamaterial structures that surpass empirically derived designs [12].

In this work, a dataset linking structural geometries to band gaps is used to train a machine learning model that can perform accurate inverse predictions, determining optimal structures for specified target band gaps.

II. Theory

A. Seismic Wave

Seismic waves are generated by the release of elastic energy during fault rupture or plate motion [13]. According to the type of vibration, seismic waves can be broadly divided into in-plane waves (P-waves) and out-of-plane waves (S-waves) [14].

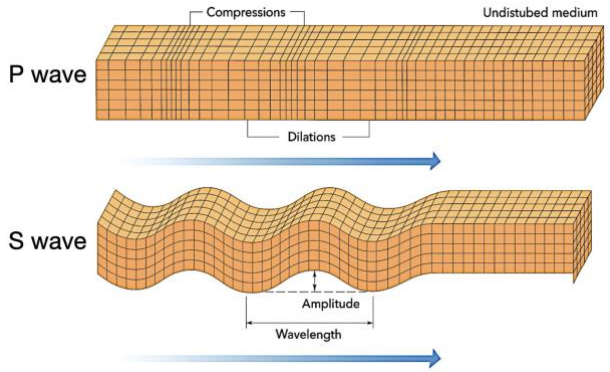

Fig.1 Comparison of P-wave and S-wave vibration modes. P wave propagates via compressions and dilations parallel to the travel direction, but S wave propagates via shear motion perpendicular to the travel direction.

P-waves are characterized by particle motion that is parallel to the direction of wave propagation, resulting in compression and rarefaction of the medium [15]. Their propagation velocity is higher than that of S-waves, and they are therefore the first seismic signals to arrive at observation stations during an earthquake [16]. However, S-waves involve particle motion perpendicular to the direction of propagation. Their velocity is slower than that of P-waves', and they cannot propagate through liquid or gaseous media [17].

To describe the propagation of both P- and S-waves in a mathematical framework, the elastic wave equation is employed. Starting from the general equation of motion in a linear elastic medium:

$$\rho \frac{\partial^2 \mathbf{u}_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} \tag{1}$$

where $\mathbf{u}_i$ denotes the displacement vector, ρ the mass density, and σ_{ij} the stress tensor, one can derive the governing equations for elastic wave propagation by introducing the strain–displacement relation and Hooke's law for isotropic materials [18]. After substitution and rearrangement, the system reduces to the well-known Navier–Cauchy equation:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u} \tag{2}$$

where λ and μ are the Lamé parameters [19].

By applying a temporal Fourier transform, the equation can be expressed in the frequency domain. In two-dimensional geometries, the displacement field separates naturally into an out-of-plane component and in-plane components [20]. This decomposition yields two distinct subsystems [21]:

1. In-plane P wave**:** the coupled in-plane displacements u_1 and u_2 satisfy

$$\nabla \cdot [\mu(r)\nabla \mathbf{u}_i(r)] + \nabla \cdot \left(\mu(r)\frac{\partial \mathbf{u}}{\partial \boldsymbol{r}_i}\right) + \frac{\partial}{\partial r_i}[\lambda(r)\nabla \cdot \mathbf{u}(r)] = -\rho(r)\omega^2 \mathbf{u}_i(r), i = 1,2 \tag{3}$$

which describe the coupled propagation of compressional and vertically polarized shear waves.

2. Out-of-plane S wave: the scalar displacement u_3 satisfies

$$\nabla \cdot [\mu(r)\nabla \mathbf{u_3}(r)] = -\rho(r)\omega^2 \mathbf{u_3}(r) \tag{4}$$

which is a scalar Helmholtz-type equation describing horizontally polarized shear waves.

Thus, beginning from the general momentum balance, the elastic wave equation can be separated into two independent classes of solutions, an uncoupled out-of-plane mode governed by a scalar equation, and a coupled in-plane system governed by vector equations. This decomposition forms the theoretical basis for analyzing seismic wave propagation in layered or heterogeneous elastic media.

B. Seismic Metamaterial

Fundamental mechanisms underlying seismic metamaterials are Bragg scattering and local resonance. In practical implementations, these metamaterials are typically realized through the periodic arrangement of engineered inclusions, commonly composed of concrete or steel, embedded in the soil [22].

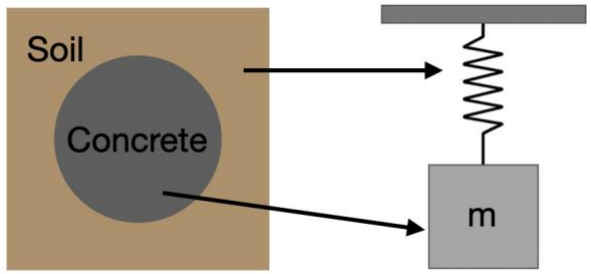

Fig. 2 Representation of the local resonance mechanism in seismic metamaterials. A concrete inclusion embedded in soil can be modeled as a spring mass system.

As shown in the figure 2, these periodic structures interact with incoming seismic waves in a manner that can be modeled by simplified spring mass system, where the soil or substrate acts as the elastic foundation and the embedded resonant bodies provide additional inertia [23]. Through this analogy, it becomes clear that local resonances can be excited within the substructures. When arranged periodically, these resonant units form elastic or phononic crystals, which display a low frequency bandgaps feature, even in the sub-wavelength regime [24]. This is particularly significant because seismic waves typically exhibit very long wavelengths, making traditional isolation methods ineffective without resorting to extremely large-scale barriers [25]. By harnessing resonance effects, seismic metamaterials provide a means to attenuate, redirect, or dissipate seismic energy more efficiently at low frequencies [26].

The bandgap characteristics of seismic metamaterials can be analyzed through elastic lattice theory [27]. A bandgap refers to a frequency range in which waves cannot propagate through any direction within the periodic medium [28].

To rigorously capture this phenomenon, the displacement field $\mathbf{u}(r)$ of the periodic structure must be expressed in terms of Bloch waves, and the governing elastic wave equation is then solved under Bloch boundary conditions, for a lattice translation vector $\mathbf{R}$,

$$\mathbf{u}(r+\mathbf{R}) = \mathbf{u}(r)e^{i\mathbf{k}\cdot\mathbf{R}} \tag{5}$$

where $\mathbf{k}$ is the wave vector, and the exponential factor accounts for the phase shift of the wavefield across lattice cells [29]. Considering the governing elastic wave equation (Eq. 3 and Eq. 4) in heterogeneous media. This formulation naturally transforms the problem into an eigenvalue analysis.

Using FEM, the displacement field can be approximated as a weighted sum of nodal values with shape functions:

$$\mathbf{u}(r) \approx \sum_i N_i(\mathbf{r})\mathbf{u}_i \tag{6}$$

where N_i are shape functions and u_i are nodal displacements. Applying Bloch boundary conditions (Eq. 5) ensures periodic consistency between adjacent cells [30]. The discretized system then takes the form of a generalized eigenvalue problem:

$$(\mathbf{K} - \omega^2\mathbf{M})\mathbf{U} = 0 \tag{7}$$

where $\mathbf{U}$ is the displacement vector, $\mathbf{K}$ is the stiffness matrix and $\mathbf{M}$ is the mass matrix [30]. The stiffness matrix arises from the discretization of the elastic strain energy and depends on elasticity modulus and Poisson's ratio. The mass matrix is derived from the kinetic energy and depends on material density and geometry. The eigenfrequencies ω obtained from Eq. (7) determine the dispersion relations. Frequency ranges lacking solutions correspond to bandgaps [31]. Thus, band gaps can be identified based on the material's Poisson's ratio, elasticity modulus, density, and geometry [32].

Solving for all $\mathbf{k}$ in 2D is impractical, so we use the Brillouin Zone, which condenses essential wavevector information [33]. In a square lattice as presented in Fig. 3, the irreducible Brillouin zone further simplifies computations by restricting analysis to a small triangular section, mirrored via symmetry. Since band gaps occur along paths Γ→X→M→Γ between high-symmetry points, analyzing only the edges suffices for band gap identification [34].

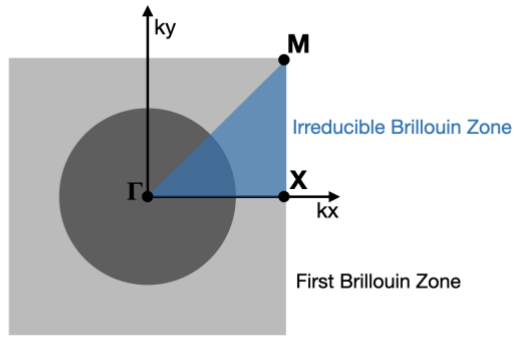

Fig. 3 First Brillouin zone and its irreducible part. The high-symmetry points Γ, X, and M are indicated along the reciprocal lattice directions.

C. *Deep learning*

Conventional bandgap design methods, such as trial-and-error, can be time-consuming and computationally expensive [35]. To address this challenge, we introduce deep learning to assist in the design process. There exists a projection relationship between the characteristic parameters of seismic metamaterials and their bandgaps [35]. By constructing an appropriate deep learning model, we can achieve forward predictions of metamaterial properties, significantly enhancing the efficiency and accuracy of the design process. In our inverse design framework, two primary machine learning architectures are employed: the multilayer perceptron (MLP) and the variational autoencoder (VAE). Together, these models form a bidirectional pipeline in which the MLP is responsible for rapid forward prediction of bandgaps, while the VAE enables inverse design, mapping target bandgap requirements back to feasible structural configurations.

The MLP is a feedforward neural network that approximates the mapping from metamaterial parameters to bandgap frequencies. Its basic operation is defined by a sequence of affine transformations followed by nonlinear activations. For an input vector $\mathbf{x}$, each layer of the MLP applies a linear transformation followed by a nonlinear activation,

$$h^{(l)} = \sigma\left(W^{(l)}h^{(l-1)} + b^{(l)}\right), l = 1, \dots, L \tag{8}$$

where $W^{(l)}$ and $b^{(l)}$ are trainable parameters and σ is typically chosen as the rectified linear unit such as ReLU [36]. The final output is obtained at the last layer, and the model parameters are optimized by minimizing a loss function such as mean squared error. This simple yet powerful structure allows the MLP to capture highly nonlinear mappings between inputs and outputs. Beyond physics and engineering, MLPs are widely used in fields ranging from finance to computer vision as universal function approximators.

The VAE is a generative model that introduces a probabilistic perspective to dimensionality reduction and data

generation [37]. It consists of an encoder, which maps input data $\mathbf{x}$ to the parameters of a latent distribution, and a decoder, which reconstructs the data from a latent vector $\mathbf{z}$. Specifically, the encoder outputs a mean $\mu(\mathbf{x})$ and variance $\sigma^2(\mathbf{x})$, defining the approximate posterior distribution,

$$q(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z};\ \mu(\mathbf{x}), \mathrm{diag}\big(\sigma^2(\mathbf{x})\big) \tag{9}$$

Here, the notation $\mathrm{diag}\big(\sigma^2(\mathbf{x})\big)$ indicates that the covariance matrix is diagonal, meaning the latent variables are assumed to be independent. Thus, each input $\mathbf{x}$ corresponds not to a single point in the latent space, but to a Gaussian distribution centered at $\mu(\mathbf{x})$ with variance $\sigma^2(\mathbf{x})$ [38].

In our project, the role of the VAE is to transform complex metamaterial geometries into low-dimensional latent representations. Compared with directly handling high-dimensional topological data, latent vectors are more compact and structured, which makes them easier for subsequent machine learning models to process and predict. In this way, intricate structural information is recast into a form that is more "intelligible" for the model, thereby enhancing the efficiency of both modeling and optimization.

It is worth noting that this idea is not limited to metamaterial design. In computer vision, VAEs have been widely applied to image generation and reconstruction [39]. After mapping a complex natural image into the latent space, the model can manipulate only a small set of latent variables to generate entirely new images. For a computer, operating directly on millions of pixels is redundant and difficult, whereas latent vectors provide a condensed and abstract representation that preserves the essential features while discarding unnecessary details [40]. Similarly, in our work, the latent vectors serve as an efficient and manageable representation of high-dimensional metamaterial geometry. Just as the VAE compresses pixels into latent variables in image generation, it compresses complex geometrical structures into latent variables in metamaterial design, enabling more effective prediction and optimization.

III. Methodology

A. Bandgap Simulation

The Finite Element Method (FEM) was employed for band gap calculations. COMSOL Multiphysics was used for specific modeling and computational analysis.

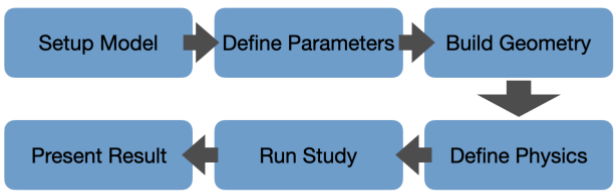

Fig. 4 Flowchart of a typical COMSOL Multiphysics simulation.

The overall workflow can be divided into six major steps. Taking the in-plane wave (P-wave) as an example, the first step is to define the basic simulation settings, including the dimensionality, the physical interface, and the study type. Here, "*2D*" was selected as the spatial dimension, the "*Solid Mechanics"* module was enabled, and an "*Eigenfrequency"* study was added in order to compute natural frequencies directly.

The second step is to specify global parameters under "*Global Definitions"*, where the density, Young's modulus, and Poisson's ratio of soil and concrete were defined, together with the wave vector as an input variable. The third step involves the construction of the geometry. The respective material properties are then assigned to each geometric region. In order to simulate an infinite periodic arrangement, Floquet-Bloch periodic boundary conditions were applied on opposing edges of the unit cell.

The fourth step is meshing, where the computational domain is discretized into finite elements to enable FEM analysis. The fifth step is to define the solution procedure. For eigenfrequency calculations, the wave vector is swept along the high-symmetry paths of the irreducible Brillouin zone, ensuring that dispersion relations are fully captured. After all parameters are set, the eigenfrequency analysis is executed to obtain the frequency response as a function of the wave vector.

Finally, the eigenfrequencies are plotted against the normalized wave vector path to generate dispersion curves. The frequency intervals without eigenmodes are identified as bandgaps. The computed data can be exported for subsequent analysis and integration into further modeling frameworks.

For out-of-plane wave (S-wave), the governing equation is not directly compatible with the formulations embedded in the *"Solid Mechanics"* module of COMSOL and therefore requires alternative physics interfaces to be employed. Due to computational cost and time considerations, out-of-plane simulations were not performed in this study. Nevertheless, it should be emphasized that the inverse design methodology presented later is equally applicable to out-of-plane waves, provided that the corresponding bandgap dataset can be obtained.

B. Dataset Generation

After obtaining the bandgaps corresponding to specific metamaterial structures through finite element simulations, the first step in training the machine learning model is to construct a sufficiently large dataset. To this end, random sampling was performed on the material parameters of the scatterers. Using the *NumPy.random* function in Python, 10,000 distinct parameter sets were generated, covering Poisson's ratio, Young's modulus, and density. The ranges of these parameters were constrained to physically feasible intervals: Poisson's ratio between 0.30 and 0.45, Young's modulus between 100 and 1000 MPa, and density between 1.5 and 2.2 g/cm³.

To generate diverse structural patterns, we adopted topological configurations to represent the basic unit cells of metamaterials. A topological configuration specifies the spatial distribution of materials within a unit, thereby capturing both geometric and compositional characteristics. During modeling, the computational domain was discretized into a grid, with each cell assigned to a material and represented by a unit vector. A finer grid provides a more accurate description of geometric features and material boundaries but comes at the expense of higher computational cost. In this study, a 50×50 grid resolution was selected to balance accuracy and efficiency.

Despite this discretization, the overall design space remains extremely large, and many randomly generated configurations are either highly redundant or physically impractical. To constrain the design space and ensure meaningful samples, we imposed the following rules:

1. Each unit cell exhibits 1/8 symmetry, reducing computational complexity and the number of unique configurations.

2. The filling fraction of scatterers is restricted to the range of 0.30–0.65, ensuring a balance between bandgap performance and material usage.
3. Scatterers must form a simply connected domain, meaning that each scatterer grid cell must share at least one edge with another, thereby preventing isolated pixels.

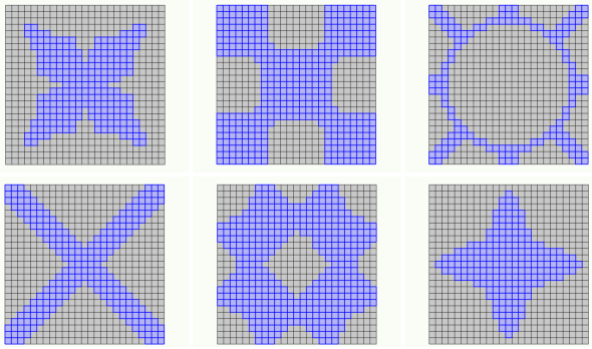

Fig. 5 The 2D geometries generated by Python based on the rules. The blue area represents the concrete, and the grey area represents the soil.

Following these rules, a total of 10,000 geometric structures were generated. The topological data were then organized into grid matrices representing the material distribution of each unit cell and exported in excel format. Using COMSOL with MATLAB scripts, material parameters were assigned to the scatterers, and the datasets were reformatted for direct import into COMSOL. with MATLAB automatically loading the datasets, running simulations, extracting results, and storing the band gap data in spreadsheets, for subsequent machine learning applications.

C. *Deep-Learning Model Construction*

Once the dataset was successfully constructed, the next step was to develop a complete deep learning model. In this work, the implementation was carried out using *TensorFlow* and *Python*. *TensorFlow* is an open-source framework developed by Google, offers several advantages, including flexible computational graph construction, support for large-scale parallelization, and strong cross-platform compatibility. These features make it particularly suitable for efficiently handling the full workflow of model design, training, and deployment [41].

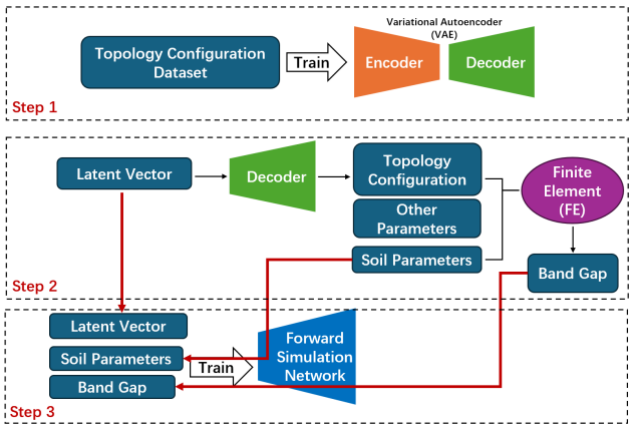

Fig. 6 Deep-learning workflow for seismic metamaterial design. A VAE encodes topology datasets into latent vectors, which are decoded and evaluated by FEM simulations. The resulting data are then used to train a forward network for rapid band gap prediction.

Firstly, let's take a wholistic view of the proposed model. As illustrated in Fig. 5, the inverse design framework for metamaterials can be divided into three major components. The central idea can be concluded as: a VAE is first employed to compress the high-dimensional geometric representation of metamaterials into a low-dimensional latent vector. The inverse design model then operates exclusively within this latent space to perform prediction and optimization. Finally, the decoded output from the VAE is used to reconstruct the corresponding metamaterial geometry [42].

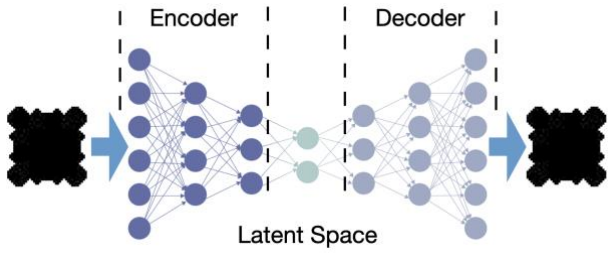

Fig. 7 The VAE encoder maps input topology data into a low-dimensional latent space, from which the decoder reconstructs the corresponding output.

In the network architecture, the encoder employs multiple VALID convolutions combined with 2×2 pooling to progressively compress the input image from 50×50 down to 1×1×128. A fully connected layer then produces a 128-dimensional latent representation, which is linearly mapped to the mean and log-variance of the latent distribution. On the decoder side, a fully connected layer first expands the latent vector into a 5×5×128 seed feature map, which is subsequently up sampled through a series of transposed convolutions, eventually reconstructing the output with dimensions 50×50×1. Throughout the network, Batch Normalization and ReLU activations are extensively applied to stabilize training and facilitate convergence. In this VAE evaluation, the reconstructed images and the ground-truth inputs are first concatenated, and both are binarized by rounding the sigmoid outputs (equivalent to a 0.5 threshold). For each sample, the code compares the two 50×50 binary images pixel-by-pixel, computes the mean of the equality mask to get the pixel accuracy, and then defines the error as [43]:

$$error = 1 - \frac{1}{N}\sum_{p=1}^{N} 1[\tilde{y}_p = y_p] \tag{10}$$

Subsequently, the latent vectors are decoded back into topological configurations, which are subjected to batch finite element simulations to obtain the corresponding bandgap results. This process yields a dataset that associates latent vectors with physical performance labels, thereby establishing the foundation for subsequent model training.

For the final step, the dataset was first constructed by reading three worksheets from an Excel file, corresponding to latent vectors, soil parameters, and bandgap results. Each column of data was normalized [43] using

$$x' = \frac{x - min}{max - min} \tag{11}$$

and the maximum and minimum values were recorded for subsequent denormalization during evaluation. The dataset was then partitioned into training, validation, and testing sets with a ratio of 8:1:1. The prediction model was implemented as a MLP with the architecture: Input → 1024 → 512 → 256 → 256 → 128 → Output (6). Batch normalization and ReLU activations were applied at each hidden layer to stabilize training, while a Sigmoid activation was used at the output layer to restrict predictions to the range [0,1], consistent with the normalized labels.

During training, a full-batch scheme was employed. The model was optimized with the Adam optimizer (learning rate $= 1 \times 10^{-3}$) and trained for 1000 epochs using the mean

squared error (MSE) loss function. Training and validation loss, along with accuracy, were recorded every 10 epochs, and model checkpoints were saved every 100 epochs. Accuracy was defined as [43]:

$$acc = 1 - mean\left(\frac{|pred - ture|}{ture}\right) \quad (12)$$

After training the forward surrogate and the VAE, the inverse design model is employed to search for latent vectors and lattice constants that correspond to a specified target bandgap, and to reconstruct the associated geometry.

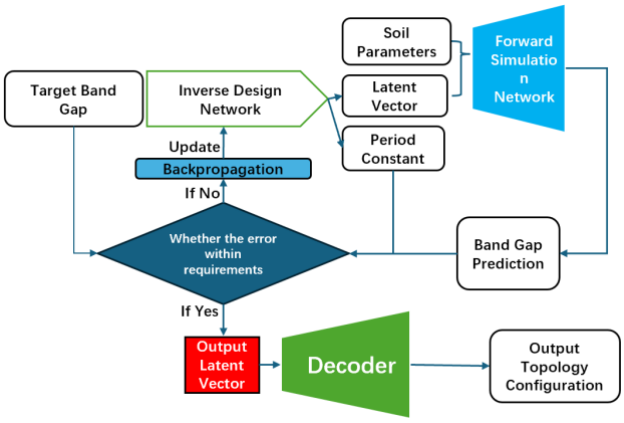

Fig. 8 Workflow of the inverse design framework. The target band gap guides the inverse design network to iteratively optimize the latent vector via backpropagation. Once the error is within tolerance, the latent vector is decoded to generate the corresponding topology configuration.

The process begins by obtaining the maximum and minimum values of bandgaps, soil parameters, and latent vectors to enable normalization and denormalization. The design objectives, including target bandgaps and soil parameters, are then extracted from the input file and fed into the inverse network.

Given the target bandgaps, the inverse network outputs normalized latent vectors and lattice constants, which are subsequently denormalized for storage and decoding. In parallel, the pretrained forward surrogate remains frozen and evaluates the candidate latent vectors together with soil parameters, producing bandgap predictions that are denormalized and scaled by the lattice constant to ensure dimensional consistency.

The inverse network is trained by minimizing the discrepancy between the predicted and target bandgaps, using the Adam optimizer with a maximum of 10,000 iterations and an early stopping criterion when the epoch exceeds 1,000 and the loss falls below a predefined threshold. Once optimized, the latent vectors are decoded by the VAE into binary geometric images.

The results including latent vectors, lattice constants, soil parameters, target bandgaps, and loss values are stored in an Excel file, with geometries also exported as images for further analysis. This procedure ensures physical consistency while enabling an automated mapping from target bandgaps to metamaterial geometries.

IV. Evaluation of Results

A. Deep-Learning Model Assessment

To validate the effectiveness of the proposed deep learning framework, this section evaluates its key components, including the reconstruction accuracy of the VAE and the prediction performance of the forward surrogate.

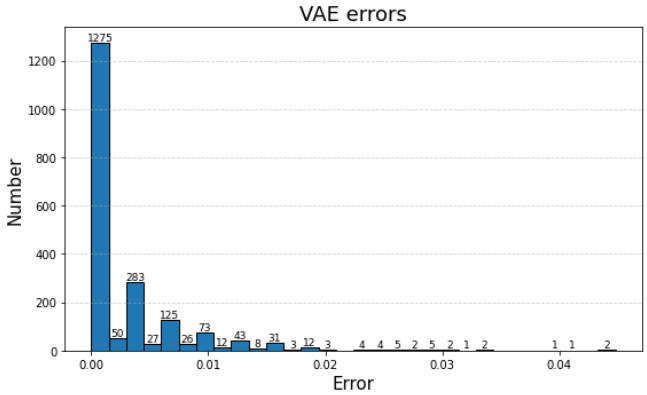

Fig. 9 Error distribution of the VAE reconstructions.

The error distribution between the original topological configurations and their reconstructions was first analyzed, as shown in Fig. 9. The results indicate that the majority of samples exhibit errors close to zero, demonstrating that the VAE can accurately reproduce the original geometries in most cases. In particular, more than one thousand samples achieved reconstruction errors below 0.005, while the number of samples decreases rapidly as the error increases. Only a few cases exhibited errors larger than 0.02, suggesting that the latent space learned by the VAE provides a stable and effective representation of complex high-dimensional structures, enabling accurate decoding.

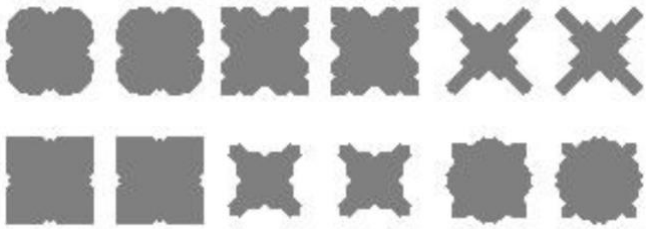

Fig. 10 Comparisons of original and reconstructed metamaterial geometries. The left column shows the original topological configurations, while the right column presents the corresponding reconstructions obtained through VAE encoding–decoding.

Representative examples of reconstructed geometries are further illustrated in Fig. 10. The left column shows the original metamaterial topologies, while the right column presents their corresponding reconstructions obtained through the VAE encoding–decoding process. It can be observed that the reconstructed geometries closely match the original ones in both overall shape and boundary details, indicating that the VAE not only preserves global structural features but also retains fine-scale local characteristics. Although minor smoothing effects can be observed at the edges of certain complex structures, the overall discrepancy is negligible and does not affect subsequent bandgap calculations.

Fig. 11 presents the evolution of loss and accuracy for the forward surrogate model during training. The horizontal axis corresponds to the number of epochs, while the vertical axis indicates the loss or accuracy values. Training results are shown as continuous lines, and validation results are represented by dots. In this study, each epoch consisted of 10 iterations of parameter updates. As illustrated, both training and validation losses decrease rapidly to near zero within the first few epochs and remain stable throughout the training process, indicating that the model does not exhibit significant overfitting or underfitting.

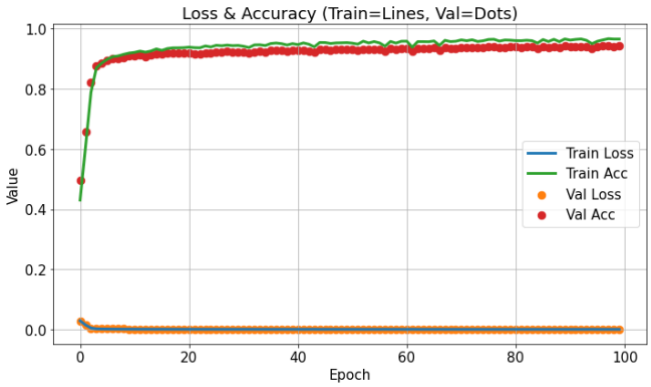

Fig.11 Training and validation curves. Blue and green solid lines represent training loss and training accuracy. Orange and red dots denote validation loss and validation accuracy.

At the same time, the accuracy of both training and validation sets exceeds 90% after approximately 10 epochs and eventually converges to about 96%. These findings demonstrate that the proposed MLP-based forward prediction network effectively captures the mapping between latent vectors and bandgaps, while maintaining strong generalization capability.

B. Case Study and Results

In the previous sections, the framework and its performance were thoroughly examined. Building upon this validated foundation, the present section applies the deep-learning model to practical scenarios.

There are three representative earthquakes selected: the 2008 Wenchuan (Mw 7.9) [44], the 1999 Chi–Chi (Mw 7.6) [45], and the 2011 Tohoku (Mw 9.0) [46] events.

EARTHQUAE INFORMATION

Event	Density (g/cm^3)	Poisson's Ratio	Young's Modulus (Mpa)	Frequency Range (Hz)
Wenchuan	1.8	0.3	650	11-18
Chi-Chi	2.7	0.25	800	20-35
Tohoku	1.65	0.24	310	5-10

Table.1 Soil parameters and dominant seismic wave frequency ranges for three major earthquakes.

The Wenchuan earthquake, with a focal depth of about 18 km, was dominated by low-frequency waves and caused widespread destruction, providing a typical scenario for examining large-scale seismic interactions. In contrast, the Chi–Chi earthquake, with its shallow depth and strong high-frequency components, serves as a case for assessing metamaterials in mid- to high-frequency mitigation. The Tohoku earthquake, as a megathrust event with broadband low-frequency content, triggered a devastating tsunami and represents a challenging testbed for complex seismic conditions. Taken together, these three events, differing in magnitude, depth, and frequency characteristics, provide a comprehensive basis for evaluating metamaterial designs across diverse seismic environments. The basic soil parameters and frequency ranges associated with these events are summarized in Table 1.

Fig. 12 illustrates the metamaterial structure designed for the Wenchuan earthquake and its corresponding band structure. Panel (a) presents the inverse-designed topological configuration with a lattice constant of 3.89 m, while panel (b) shows the calculated band diagram. A clear bandgap is observed between 11.204 Hz and 17.775 Hz. Compared with the target frequency range of 11–18 Hz, the error are

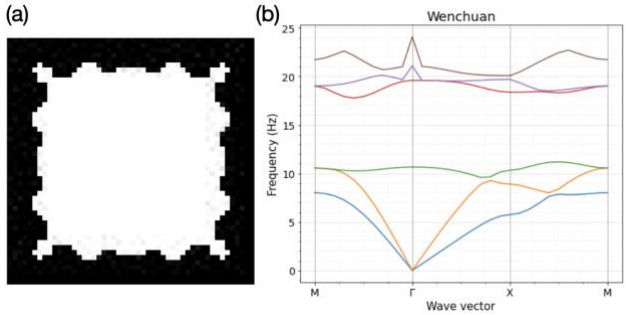

Fig.12 (a) Designed seismic metamaterial structure for the Wenchuan case. (b) The corresponding energy band diagram for the Wenchuan case.

approximately 1.85%, indicating an excellent agreement. The loss value converges to 0.01, further confirming the high accuracy of the prediction. Notably, due to the dominance of low-frequency components in the Wenchuan earthquake, the designed lattice constant is relatively large to ensure that the bandgap adequately covers the target range.

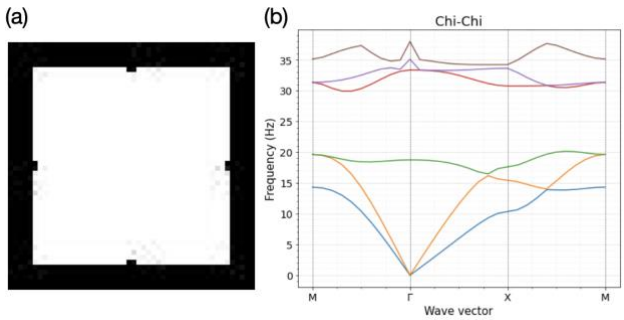

Fig.13 (a) Designed seismic metamaterial structure for the Chi-Chi case. (b) The corresponding energy band diagram for the Chi-Chi case.

Fig. 13 presents the metamaterial structure designed for the 1999 Chi–Chi earthquake and its corresponding band diagram. Panel (a) shows the inverse-designed geometry with a lattice constant of 2.21 m, while panel (b) illustrates the resulting band structure. The model converged to a loss value of 0.03, indicating a reliable prediction. The computed bandgap extends from 21.14 Hz to 31.9 Hz. Compared with the target range of 20–35 Hz, the lower boundary aligns well, but the upper boundary falls short of fully covering the 35 Hz target. This discrepancy arises from the fact that the target frequency band is unusually wide, making it challenging for the current metamaterial framework to fully capture the higher-frequency range.

It is also notable that the filling ratio of this structure has reached the imposed upper limit of 0.65, suggesting that the material capacity is already maximized. While the overall bandgap coverage is limited in the higher-frequency portion, partial bandgaps can still be observed around common incident directions, particularly near the Γ point, where frequencies extend up to 33.324 Hz. This demonstrates that, despite incomplete coverage at the upper boundary, the model remains capable of providing meaningful vibration isolation at mid- to high-frequency ranges, confirming the feasibility and potential of the proposed framework for such seismic conditions.

Fig. 14 presents the metamaterial unit structure designed for the 2011 Tohoku earthquake and its associated band structure. Panel (a) shows the inverse-designed geometry, where the lattice constant was optimized to 4.0 m, while panel (b) illustrates the computed band diagram. The model converged to a loss value of 0.52, which is notably higher

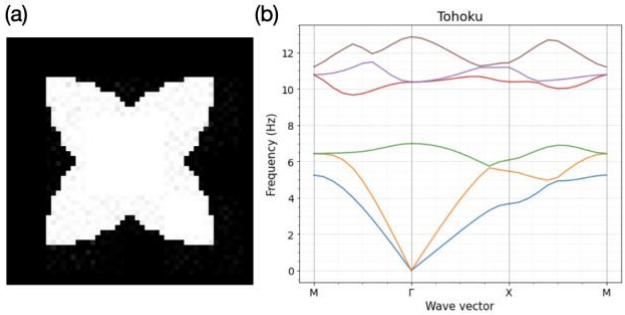

Fig.14 (a) Designed seismic metamaterial structure for the Tohoku case. (b) The corresponding energy band diagram for the Tohoku case.

than in the previous cases but still within an acceptable range. The simulation reveals a bandgap between 6.987 Hz and 9.655 Hz. Compared with the target range of 5–10 Hz, the designed bandgap successfully covers most of the intended interval, particularly aligning well with the upper boundary.

It is important to highlight that the lattice constant has already reached the imposed maximum of 4.0 m, making it challenging to further lower the bandgap edge toward the target's lower bound. This constraint likely contributes to the higher loss value observed in this case.

V. Conclusion and Discussion

In this work, we proposed an integrated framework that combines physics-based simulations and deep learning to enable the inverse design of seismic metamaterials. The underlying concept originates from band theory in periodic structures and the mechanism of local resonance, which together provide the physical foundation for achieving bandgap control in the low-frequency, long-wavelength regime. However, conventional forward design approaches typically rely on repetitive parameter sweeps and extensive numerical simulations, which are highly inefficient and often struggle to identify optimal solutions within the vast space of structural and material configurations. To address this limitation, we first formulated the governing elastic wave equations and employed the finite element method (FEM) in COMSOL Multiphysics to compute band structures under various soil and structural parameters.

To overcome the complexity of high-dimensional geometric representations, we introduced a VAE to compress metamaterial topologies into low-dimensional latent vectors while ensuring reconstruction fidelity. These latent variables provided a compact yet informative design space for learning. Based on this representation, a MLP surrogate model was trained to predict bandgaps from latent vectors and soil parameters, drastically reducing computational cost compared with FEM. Finally, an inverse design network was developed to iteratively optimize latent vectors toward matching target bandgaps. The optimized latent vectors were decoded back into geometries and revalidated through FEM, establishing a closed-loop workflow that bridges physical simulations and machine-learning-assisted design.

The proposed framework was then applied to three representative earthquake scenarios, including the 2008 Wenchuan, 1999 Chi–Chi, and 2011 Tohoku events. Case studies demonstrated that the designed metamaterial structures could generate bandgaps closely aligned with the target seismic frequency ranges, achieving low prediction errors and reasonable coverage despite variations in frequency content and design constraints. These findings confirm the effectiveness and practicality of integrating physics-based simulations with deep learning for the systematic design of seismic metamaterials. The results highlight the potential of such hybrid approaches to support seismic hazard mitigation by enabling efficient, accurate, and scalable design of metamaterial-based isolation systems.

While the proposed framework demonstrates promising results, several limitations remain and point toward future directions. First, due to the constraints of both available data and computational resources, the present study focused exclusively on in-plane P-wave analysis. Nevertheless, the methodology itself is general and can be directly extended to S-waves when resources permit. In future work, we aim to construct a design framework that simultaneously addresses both wave types, thereby providing more comprehensive seismic protection.

Second, the use of binary materials, specifically soil and concrete, introduces inherent limitations. The relatively small contrast in their elastic properties restricts the strength of local resonance effects and narrows the achievable bandgap range. To overcome this limitation, future studies may incorporate additional material types to enhance resonance. For instance, previous seismic metamaterial studies have demonstrated that embedding rubber layers between soil and concrete can amplify local resonance and produce wider and lower-frequency bandgaps. Extending our framework to accommodate multi-material systems would therefore be a natural next step.

Finally, the current work was restricted to two-dimensional structures. Although three-dimensional metamaterials are expected to exhibit superior performance, their inverse design is significantly more challenging due to prohibitive computational costs. In particular, a direct extension of the present voxel- or pixel-based discretization approach to 3D would result in an exponential growth of computational complexity. This highlights the need for more advanced algorithms or dimensionality reduction strategies to make 3D inverse design feasible. We believe that exploring such directions could further unlock the potential of seismic metamaterials for practical large-scale applications.

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